P08

Byzantine Fault Tolerance in Self-Healing Peer-to-Peer Meshes

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Domain DOM_CS

Abstract

Practical Byzantine Fault Tolerance guarantees state machine replication and liveness across asynchronous networks containing corrupt or malicious nodes. By maintaining cryptographic digests and quorum proofs, peer meshes resist coordinated subversion.

Abstract quoted from M. Castro et al., ACM TOCS (2002).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?